



**KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY
(AN AUTONOMOUS INSTITUTION)**

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SDC1-WEEK1-Day2

II-I CSE, CSE(AI&ML) AY: 2026-2027

SL No	Week	Part	Detailed Subtopics	Part	Detailed Subtopics
1	Week 1	Part A	Overfitting: Causes & Solutions, Regularization (L1 - Lasso, L2 - Ridge)	Part B	Evaluation metrics

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(Week 1-Session 2 Part B: Evaluation metrics)

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1. Why Evaluation Metrics Matter

Training a model is only half the task — the other half is measuring how well it actually performs.

Evaluation metrics quantify the gap between a model's predictions and the true outcomes, and that number is what guides model selection, hyperparameter tuning, and the decision to deploy a model or not.

Different metrics highlight different kinds of mistakes. A metric that looks reassuring (like high accuracy) can hide a model that is practically useless, so choosing the **right** metric for the problem matters as much as building the model itself.

Regression and Classification problems are evaluated differently, since their outputs are fundamentally different: regression predicts a continuous number, while classification predicts a discrete category.

2. Regression Evaluation Metrics – Overview

Regression models predict continuous numeric values — price, temperature, salary, yield, and so on. Their quality is measured by **how close the predicted values are to the actual values**. The commonly used regression metrics are:

- **MSE (Mean Squared Error)** — average of squared errors
- **MAE (Mean Absolute Error)** — average of absolute errors
- **RMSE (Root Mean Squared Error)** — square root of MSE, back in the original units
- **R² (Coefficient of Determination)** — proportion of variance explained by the model
- **Adjusted R²** — R² adjusted for the number of predictors used

Each is examined below, with the situations where it is the right choice.

3. Mean Squared Error (MSE)

MSE is the average of the squared differences between actual and predicted values.

$$\text{MSE} = (1/n) \times \sum (y_i - \hat{y}_i)^2$$

n = number of data points

y_i = actual value

$\hat{y}_i$ (*y-cap*) = predicted value

Properties

- Squaring the error means **large errors are penalized much more heavily** than small ones
- Differentiable, which makes it well suited for optimization methods such as gradient descent
- **Highly sensitive to outliers** — a few very large errors can dominate the score
- Result is in **squared units** of the target variable, which makes it harder to interpret directly

Example

Actual values: 50, 60, 70. Predicted values: 55, 58, 75.

Errors: 5, -2, 5 → Squared errors: 25, 4, 25 → MSE = (25 + 4 + 25) / 3 = 18.

4. Mean Absolute Error (MAE)

MAE is the average of the absolute differences between actual and predicted values.

$$\text{MAE} = (1/n) \times \sum |y_i - \hat{y}_i|$$

n = number of data points
 y_i = actual value
 $\hat{y}_i$ (y -cap) = predicted value

Properties

- Treats every error with **equal weight** — no exaggeration of large errors
- **Robust to outliers** compared to MSE and RMSE
- Same unit as the target variable, so it is easy to explain: “the model is off by about X units on average”

Example

Using the same values as above: $|5| + |-2| + |5| = 12 \rightarrow \text{MAE} = 12 / 3 = 4$.

5. Root Mean Squared Error (RMSE)

RMSE is the square root of MSE, which brings the error back into the original unit of the target variable.

$$\text{RMSE} = \sqrt{[(1/n) \times \sum (y_i - \hat{y}_i)^2]}$$

n = number of data points
 y_i = actual value
 $\hat{y}_i$ (y -cap) = predicted value

Properties

- Still penalizes large errors more than small ones (inherited from MSE), but is more interpretable since it is in the original units
- A balanced choice between MAE (simple, robust) and MSE (sensitive, harder to interpret)

Example

From the MSE example above, $\text{MSE} = 18 \rightarrow \text{RMSE} = \sqrt{18} \approx 4.24$.

6. R^2 – Coefficient of Determination

R^2 measures the proportion of variance in the target variable that the model is able to explain, relative to simply predicting the mean every time.

$$R^2 = 1 - (\text{RSS} / \text{TSS})$$

$\text{RSS} = \sum (y_i - \hat{y}_i)^2$ (residual sum of squares)
 $\text{TSS} = \sum (y_i - \bar{y})^2$ (total sum of squares)
 $\bar{y}$ (y -bar) = mean of actual values

Properties

- Ranges from 0 to 1 for a reasonable model; a value close to 1 means the model explains most of the variance
- Can go **negative** if the model performs worse than simply predicting the average every time
- Says nothing about the **size** of the error, only how much variance is explained — useful for comparing model fit, not for reporting typical error magnitude

7. Adjusted R²

Plain R² has a weakness: it **never decreases** when more predictors are added, even irrelevant ones.

Adjusted R² corrects for this by penalizing the addition of predictors that do not meaningfully improve the model.

$$\text{Adj. } R^2 = 1 - [(1 - R^2)(n - 1) / (n - k - 1)]$$

n = number of observations

k = number of predictors in the model

R² = coefficient of determination

Properties

- Increases only if a new predictor genuinely improves the model more than would be expected by chance
- Preferred over plain R² when comparing **multiple regression models with different numbers of predictors**
- Helps guard against overfitting caused by adding unnecessary features

8. Choosing the Right Regression Metric

Metric	Sensitive to Outliers	Same Unit as Target	Best Used When
MSE	High	No (squared units)	Large errors must be penalized heavily; used internally for optimization
MAE	Low	Yes	A simple, robust “average mistake” figure is needed
RMSE	Medium–High	Yes	A balance between interpretability and penalizing large errors
R ²	—	— (proportion)	Comparing how well models explain the variance in the data
Adjusted R ²	—	— (proportion)	Comparing models that use different numbers of predictors

9. Classification Evaluation Metrics – Overview

Classification models predict a **discrete category** — spam or not spam, disease or healthy, fraud or legitimate. Evaluating them starts by comparing predicted labels against actual labels, which is summarized in a **Confusion Matrix**. From this matrix, several metrics can be derived: Accuracy, Precision, Recall, F1 Score, and the ROC-AUC curve.

10. Confusion Matrix

A **Confusion Matrix** is a table that shows the number of correct and incorrect predictions, broken down by actual versus predicted class. For binary classification it looks like this:

	Predicted Positive	Predicted Negative
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

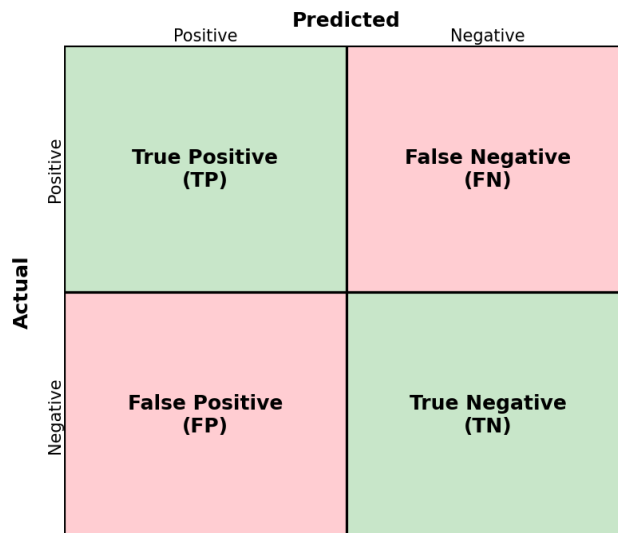


Figure: Confusion matrix layout for binary classification

Definitions

- **True Positive (TP):** Predicted positive, and it actually is positive.
- **False Positive (FP):** Predicted positive, but it is actually negative. (Type I error)
- **True Negative (TN):** Predicted negative, and it actually is negative.
- **False Negative (FN):** Predicted negative, but it is actually positive. (Type II error)

11. Accuracy

Accuracy measures the overall proportion of correct predictions.

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$$

TP, TN = correct predictions
FP, FN = incorrect predictions

When to Use

Accuracy is a fair summary **only when classes are roughly balanced** (similar numbers of positive and negative examples).

Caution: Misleading with Imbalanced Classes

Disease detection dataset: 950 healthy, 50 sick. A model that predicts “healthy” for **every** patient still scores 95% accuracy — despite catching zero actual cases of disease. Accuracy alone hides this failure.

12. Precision

Precision measures how many of the model's **positive predictions were actually correct**. It is also called the Positive Predictive Value.

$$\text{Precision} = TP / (TP + FP)$$

TP = correctly predicted positives
FP = incorrectly predicted positives

When to Use

When **false positives are costly**. Example: spam detection — marking a genuine email as spam can bury important messages, so precision should be high.

13. Recall

Recall measures how many of the **actual positives the model successfully identified**. It is also called Sensitivity or the True Positive Rate.

$$\text{Recall} = TP / (TP + FN)$$

TP = correctly predicted positives
FN = actual positives the model missed

When to Use

When **false negatives are costly**. Example: cancer detection — missing an actual case is dangerous, so recall should be high, even if it means a few extra false alarms.

14. F1 Score

The F1 Score is the **harmonic mean** of Precision and Recall. It is useful when both false positives and false negatives matter, and a single balanced number is needed.

$$F1 = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

Precision = $TP / (TP + FP)$
Recall = $TP / (TP + FN)$

Because it is a harmonic mean rather than a simple average, F1 stays low if *either* Precision or Recall is low — a model cannot get a good F1 score by excelling at only one of the two.

15. ROC Curve and AUC

The **ROC (Receiver Operating Characteristic) curve** plots the True Positive Rate against the False Positive Rate at every possible classification threshold, rather than a single fixed threshold like 0.5.

$$TPR = TP / (TP + FN)$$

$$FPR = FP / (FP + TN)$$

TPR = True Positive Rate (Recall)
FPR = False Positive Rate

The **AUC (Area Under the Curve)** summarizes the entire ROC curve into a single number between 0 and 1, representing the model's ability to distinguish between the two classes across all thresholds.

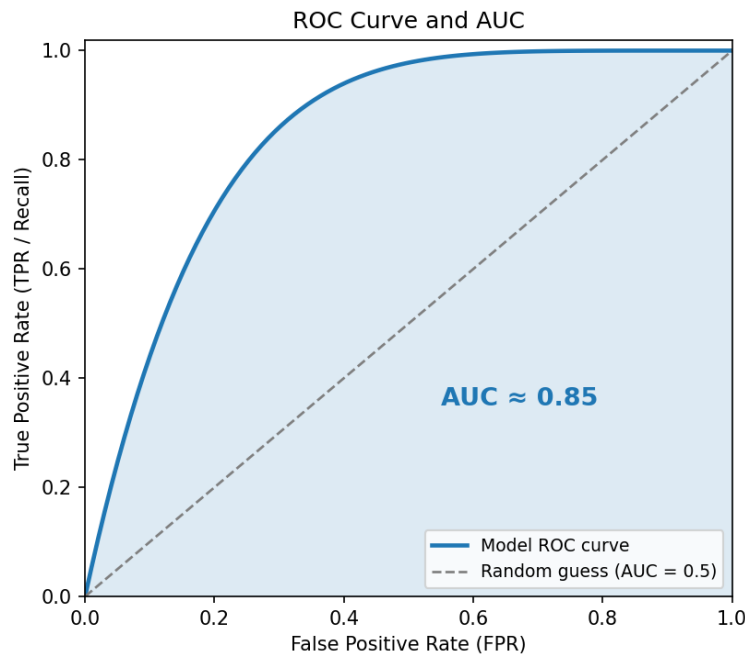


Figure: ROC curve — the further the curve bows toward the top-left corner (away from the diagonal), the higher the AUC and the better the model separates the two classes.

Interpreting AUC

- AUC = 0.5 → no better than random guessing (the diagonal line)
- AUC close to 1.0 → the model separates the two classes very well
- AUC is **threshold-independent**, which makes it useful for comparing classifiers regardless of the specific cutoff chosen

16. Choosing the Right Classification Metric

Metric	Focus	When It Is Important
Accuracy	Overall correctness	Classes are balanced
Precision	Correctness of positive predictions	False positives are costly
Recall	Capturing actual positives	False negatives are costly
F1 Score	Balance of precision and recall	Both FP and FN matter, one balanced number is needed
ROC-AUC	Threshold-independent separability	Comparing classifiers, especially with imbalanced data

Worked Example: Fraud Detection

Class imbalance: 990 legitimate transactions, 10 fraudulent. Model gets TP = 5, FP = 5, FN = 5, TN = 985.

Accuracy = $(5 + 985) / 1000 = 99\%$ — looks excellent, but the model missed half of all actual frauds.

Precision = $5 / (5 + 5) = 50\%$. Recall = $5 / (5 + 5) = 50\%$.

Since missing a fraud is costly, Recall (and AUC-ROC for overall discrimination) matters far more here than the impressive-looking 99% accuracy.

17. Summary

- Regression metrics compare predicted numbers to actual numbers: **MSE** and **RMSE** penalize large errors more, **MAE** is robust and simple to interpret, **R²** and **Adjusted R²** measure explained variance.
- Classification metrics are derived from the **Confusion Matrix**: **Accuracy** for balanced classes, **Precision** when false positives are costly, **Recall** when false negatives are costly, **F1** for a balance of both, and **ROC-AUC** for threshold-independent comparison.
- No single metric tells the whole story — the right metric always depends on which type of mistake is more costly for the specific problem.

Q & A

1. Why is accuracy sometimes a misleading metric?

Answer:

Accuracy can look high simply because one class dominates the dataset (class imbalance). A model that always predicts the majority class can score high accuracy while completely failing to identify the minority class, which is often the class that matters most.

2. What is the key difference between Precision and Recall?

Answer:

Precision measures how many of the predicted positives were actually correct ($TP / (TP+FP)$), focusing on false positives. Recall measures how many of the actual positives were correctly identified ($TP / (TP+FN)$), focusing on false negatives. They highlight different types of mistakes.

3. Why does MSE penalize large errors more than MAE?

Answer:

MSE squares each error before averaging, so a large error contributes disproportionately more to the total than several small errors. MAE takes the absolute value, so every unit of error contributes equally regardless of size.

4. What does an R² value of 0 or a negative value mean?

Answer:

An R² of 0 means the model explains no more variance than simply predicting the mean every time. A negative R² means the model performs worse than that simple baseline — a sign of a poorly fitted model.

5. Why is AUC described as “threshold-independent”?

Answer:

AUC is computed from the ROC curve, which plots TPR against FPR across every possible classification threshold, not just one fixed cutoff (like 0.5). This makes AUC a summary of separability across all thresholds, rather than a judgment tied to one specific decision boundary.

Scenario-Based Questions**6. Scenario-Based Question****Scenario:**

A bank wants to predict credit card spending limits. A few predictions are very far from the actual values, and the bank wants a metric that penalizes large errors more heavily than small ones.

Question:

Which regression metric should be used, and why?

Answer:

MSE (Mean Squared Error) should be used. Squaring the errors means large deviations grow disproportionately, which is exactly the behaviour the bank wants. MSE is also differentiable, which makes it convenient for model optimization.

7. Scenario-Based Question**Scenario:**

A sales team wants to know, on average, how far off their demand forecasts are, expressed in the same units as sales figures, without the number being skewed by a handful of unusual outliers.

Question:

Which regression metric fits this requirement best?

Answer:

MAE (Mean Absolute Error) fits best. It treats every error equally (no exaggeration of large errors), stays in the original units, and is easy to explain as “on average, the forecast is off by X units.”

8. Scenario-Based Question**Scenario:**

A hospital builds a model to detect a rare disease. The model predicts every patient as healthy.

Question:

What is misleading about reporting this model's accuracy, and what should be reported instead?

Answer:

Because the disease is rare, most patients genuinely are healthy, so predicting “healthy” for everyone still produces high accuracy — despite the model catching zero actual cases. Recall, Precision, and F1 should be reported instead, since they expose this failure to detect the positive (disease) class.

9. Scenario-Based Question

Scenario:

A spam filter occasionally marks genuine, important emails as spam, which frustrates users.

Question:

Which metric should be prioritized when tuning this filter, and why?

Answer:

Precision should be prioritized. High precision means that when the filter flags an email as spam, it is very likely to actually be spam, which minimizes the annoying false positives (genuine emails wrongly marked as spam).

10. Scenario-Based Question

Scenario:

A factory quality-control model reports the following results on a batch of products: TP = 50, FP = 10, FN = 40, TN = 900. The goal is to catch as many faulty products as possible before they ship.

Question:

Which metric should be prioritized, and what does the accuracy figure hide in this case?

Answer:

Recall should be prioritized, since FN = 40 means the model is missing a large number of actual faulty products — exactly what the factory cannot afford. Accuracy here is $(50+900)/1000 = 95\%$, which looks strong but hides the fact that 40 out of 90 actual faults (nearly half) went undetected.